

Memory Banking in 8086

Q. What is Memory Banking in the 8086 microprocessor Explain the roles of the A0 and BHE# signals in accessing Even and Odd memory banks for byte and word transfers.

[DEF] Definition to Remember

Memory Banking is the hardware organization of the 8086's 1 MB physical memory into two parallel 512 KB halves: the Even Bank (connected to D0-D7) and the Odd Bank (connected to D8-D15). This allows the CPU to efficiently read/write either 8-bit bytes or 16-bit words.

1. Concept of Memory Banking

To support a 16-bit data bus while maintaining byte-level addressing, memory is split:

2. Role of A0 and BHE# Signals

The CPU uses two specific signals to select which bank to activate during a memory cycle:

3. Byte and Word Access Scenarios

BHE#	A0	Selection	Transfer Type
0	**0**	Both Banks	**Read/Write 16-bit Word from an Even
0	**1**	Odd Bank	**Read/Write 8-bit Byte from an Odd Ad
1	**0**	Even Bank	**Read/Write 8-bit Byte from an Even A
1	**1**	None	No memory access.

4. Accessing a Word from an Odd Address (Unaligned Access)

If the CPU tries to read a 16-bit word starting at an odd address (e.g., 00001H), it cannot do it in one cycle. It requires two clock cycles:

[MUST-WRITE] Must-Write Points (for 10 marks)

- Memory Banking splits 1 MB memory into two 512 KB banks: Even and Odd.
- Even Bank: Even addresses, connected to lower data bus (D0-D7).
- Odd Bank: Odd addresses, connected to upper data bus (D8-D15).
- A0=0 enables the Even Bank; BHE#=0 enables the Odd Bank.
- BHE#=0, A0=0: Reads a 16-bit word from an Even address in 1 clock cycle.
- Reading an 8-bit byte from Even sets A0=0, BHE#=1 (and vice versa for Odd).
- Reading a 16-bit word from an Odd Address (unaligned) takes 2 clock cycles.

[RECALL] Quick Recall

Memory Banking -> 2 Banks (Even/Odd) -> Even Bank (D0-D7, A0=0) -> Odd Bank (D8-D15, BHE#=0) -> Both=0 = 16-bit transfer in 1 cycle -> Unaligned Word = 2 cycles